Alessandro Bombarda

+41 772 89 49 87 | alebombarda@icloud.com | linkedin.com/in/alessandrobombarda | github.com/ale18V

Education

École Polytechnique Fédérale de Lausanne (EPFL) - Master in Computer Science (GPA: 5.7/6) Sep 2025 - Sep 2027

- Admitted with Excellence Fellowship (1% master students)

University of Bologna - Bachelor in Computer Engineering (GPA: 29.38/30) Sep 2021 - Nov 2024

- Graduated with 110/110 cum laude and awarded 2x merit scholarships

University of California Riverside - Exchange Student (GPA: 3.95/4.0) Jan 2024 - Jul 2024

Work Experience

Incoming Software Engineer Intern Jul 2025 - Sep 2025
Bending Spoons *Milan, Italy*

Research Engineer Apr 2025 - Nov 2025
Asclepyus *Remote*

- Led the development of a federated learning platform for hospitals sponsored by the European Union.
- Architechted and delivered the platform from zero to production-ready, implementing a microservices architecture using RabbitMQ, gRPC, and REST APIs.
- Integrated observability using Grafana, Prometheus, and Loki to track training performance and system health across the network.
- Established the end-to-end CI/CD pipeline, using GitHub Actions to deploy the services and publish the docker containers to GHCR.

Full Stack Engineer Aug 2024 - Feb 2025
Polarity *Reggio Emilia, Italy*

- Built a React Native mobile app for a non-custodial cryptocurrency wallet with OTC trades and on-chain payments.
- Developed a React frontend for a cryptocurrency payment processor, including trade and swap order flows.
- Coordinated with clients to refine functional and design requirements.

Projects & Extracurricular

Blockchain-Based Federated Learning System | Bachelor Thesis *github.com/ale18V/FedChain*

- Researched a proof-of-stake blockchain for federated learning, addressing performance and security flaws found in past literature.
- Developed the decentralized system using Python - implemented the peer-to-peer network using bidirectional gRPC, engineered a PoS consensus resilient to malicious updates, and secured integrity of messages with asymmetric cryptography.
- Trained models on the blockchain using PyTorch and demonstrated improved security over vanilla federated learning.

Adaptive Decentralized Learning | Research Project *github.com/ale18V/BanditDL*

- Researched adaptive sampling strategies to improve communication efficiency and convergence speed in decentralized learning, using multi-armed bandit algorithms to sample neighbors.
- Optimized the decentralized learning simulator to run on high performance clusters, improving the efficiency by 12.7x and enabling large-scale experiments.

Apertus Post-Training | Swiss AI Initiative *https://apertvs.ai/*

- Created a Reinforcement Learning Gym aimed to overcome limitations in basic vision tasks (BlindTest benchmark) and improved Apertus' performance by +33% on the specific benchmark and an average +24% across general vision benchmarks.

Parkinson's Disease Detector | EPFL LiGHT Laboratory

- Designed a transformer-based audio classification pipeline for Parkinson's disease detection from vowel phonation, achieving ~75% ROC-AUC on the mPower dataset, outperforming the CNN baseline by 15%.

Qwen Math | NLP Class

- Finetuned Qwen3-1.7B on Math problems using both SFT and RL (GRPO + DAPO) to achieve +27% averaged across AIME24/25 and AMC23 (ranked 1st in class).

Cybersecurity Tutor and Competitor | CyberChallenge.IT & polygl0ts

- Qualified for University of Bologna team in the national program CyberChallenge.IT 2023; secured top 8 in the final competition by managing team infrastructure and exploiting/patching security vulnerabilities of cryptographic protocols and web applications.
- Co-organized PascalCTF25/26, UlisseCTF25 and LakeCTF25; tutored students for the Italian cybersecurity competition OliCyber.IT.
- Participate in international cybersecurity competitions with UliSSE (University of Bologna) and polygl0ts (EPFL).

Skills

Programming Languages: Go, Python, Java, C/C++, JavaScript, TypeScript, PHP, SQL, Bash, Solidity, Move.
Tools and Frameworks: Git, Github, Linux, Docker, Web3, FastAPI, Flask, React, Vite, React-Native, OpenAPI, gRPC, MySQL.
Expertise: Web3, Blockchain, Machine Learning & AI, Cybersecurity & Penetration Testing, Competitive Programming, Compilers, Operating Systems, Network Protocols, Teamwork, Leadership, Critical Thinking.
Languages: English (C1), Italian (Native), French (A2)